SELECTION OF SAMPLE

Ch#5

Example: All higher institution in Gujranwala Division. (25 To 30)

5.1 POPULATION:

Population is the group of interest for the researcher. This group has at least one characteristic that differentiates it from others. This is the group to whom researcher generalizes the result.

(1 institution or 4-5 institution).

5.2 SAMPLE: Name.

A sample is a small proportion of a population selected by specific procedures for observation and analysis. The characteristics of the sample represent the characteristics of population from which it is drawn.

Process.

5.3 SAMPLING:

Sampling is a process of selecting a number of individual for a study in such a way that the individuals represent the larger group from which they were selected.

5.4 SAMPLING TECHNIQUES:

Probability Sampling: Equal Opportunity for sample.

A probability sampling is that in which the sample is selected in such a way that every individual of a population has a known chance / probability of being included in sample.

Non-Probability Sampling: Due To CGPA, Height Age, City / Village.

In this sampling, it is not possible to specify the probability or chance that each member of a population has of being selected for the sample. Certain crieteria

5.5 TYPES OF PROBABILITY SAMPLING:

Most simple and easy Technique

5.5.1 RANDOM SAMPLING: Equal Chance. No Random No Criteria: classification

Random sampling is the process of selecting a sample in such a way that all individuals in the defined population have an equal and independent chance of being selected for the sample. Example: Gujranwala District, Division, Specific Area

Steps:

1. Identify population.	5000 teachers
2. Determine sample size (10%)	500 teachers
3. List all teachers.	List of 5000 teachers
4. Assign Individuals consecutive no.	Assign no. from 0000-4999

59058
11859
53634
48708

7 1 7 1 0
8 3 9 4 2
3 3 2 7 8

5. Select a no. in random from random table. 53634
6. Look at appropriate No. of digits last four digits (3634) (as population is 5000)
7. If no. is in list then select (3634) ok in sample.
8. Go to next no. 48708 last four digits (8708) not in list - ignored.
9. Next Nos. last four digits 1710, 3942, 3278 Included in Sample
10. Carry on until selection of 500 teachers.

Make Sub-Group → P E H
Select Sub Group.

5.5.2 STRATIFIED SAMPLING:

Jis group ko sel kry gaye wo us ka hi representation

Sampling in which sub-groups in population "have representation in sample" in the same proportion as they exist in population. Sample shows the characteristics of population

Steps:

1. Identify population. 5000 teachers
2. Determine sample size (10%) 500 teachers
3. Identify sub groups
   (a) Primary Teachers
   (b) Elementary Teachers
   (c) High School
4. Classify all members of population.

Primary teachers $= 2500 = \frac{2500}{5000} \times 100 = 50\%$

Elementary teachers $= 1500 = \frac{1500}{5000} \times 100 = 30\%$

High Schools $= 1000 = \frac{1000}{5000} \times 100 = 20\%$

5. Proportionate representation

Primary Teachers $= 50\% = \frac{50}{100} \times 500 = 250$

Elementary Teachers $= 30\% = \frac{30}{100} \times 500 = 150$

High Teachers $= 20\% = \frac{20}{100} \times 500 = 100$

6. Use a table of random sample and select

Primary teachers	250
Elementary teacher	150
High teachers	100
Total	500

School/Group.

5.5.3 CLUSTER SAMPLING: Select whole group

Sampling in which groups/ institutions are randomly select rather than individual.

Steps:

1. Identify population 5000 teachers
2. Determine sample size10% 500 teachers
3. Identify cluster School
4. List all clusters (School/Institutions.) 100 schools
5. Estimate average no. of members population per cluster 50 teachers of
6. No. of school needed $= \frac{\text{Desired sample size}}{\text{Average cluster}} = \frac{500}{50} = 10$

Collect data from all group.

few cases in order to ensure the depth and authenticity of supplied information or interview may be arranged for those cases who have failed to answer the questionnaire. Thus double sampling may help towards deeper information authenticity of information and completeness of information. In the usual double sampling design, the investigator selects his sample on the basis of a characteristic which is readily available and highly correlated with the primary characteristics for which the collection of data is expensive or difficult. Since the two characteristic are correlated an adequate sample with respect to the second characteristic should automatically also be an adequate sample with respect to the first. However one sample is analyzed and information obtained is used to draw the next sample to examine the problem further.

Every one has no equal

5.6 TYPES OF NON-PROBABILITY SAMPLING:

Chance of smapling.

5.6.1 CONVENIENCE SAMPLING:

Conveient Available. person.

Convenience sampling, also referred to as accidental sampling or haphazard sampling, is the process of including whoever happens to be available at the time. Two examples of convenience sampling are seeking volunteers and studying existing groups "just because they are there.") For example, you've been part of a convenience sample if you have been stopped on the street or in a grocery store by someone who wants your opinion of an event. Those who volunteer to answer are usually different from nonvolunteers, and they may be more motivated or more interested in the particular study. Because the total population is composed of both volunteers and nonvolunteers, the results of a study based solely on volunteers are not likely generalizable to the entire population. Suppose you send a questionnaire to 100 randomly selected people and ask the question, "How do you feel about questionnaires"? Suppose that 40 people respond, and all 40 indicate that they love questionnaires. Should you then conclude that the entire population loves questionnaire? Certainly not. The 60 who did not respond may not have done so simply because they hate questionnaires?

Have some criteria. Select those person who have desired characteristics

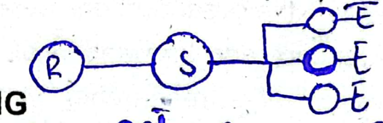
5.6.2 PURPOSIVE SAMPLING

Purposive sampling, also referred to as judgment sampling, is the process of selecting a sample that is believed to be representative of a given population. In other words, the researcher selects the sample using his experience and knowledge of the group to be sampled. For example, if a researcher plans to study exceptional high schools, he can choose schools to study based on his knowledge of exceptional schools. Prior knowledge or experience can lead the researcher to select exceptional high schools that meet certain criteria, such as a high proportion of students going to four-year colleges, a large number of Advanced Placement students, extensive computer facilities, and a high proportion of teachers with advanced degrees. Notice the important difference between convenience sampling, in which participants who happen to be available are chosen, and purposive sampling, in which the researcher deliberately identifies criteria for selecting the sample. Clear criteria provide a basis for describing and defending purposive samples, and as a result, the main weakness of purposive sampling is the potential for inaccuracy in the researcher's criteria and resulting sample selections.

when we have large population.

5.6.3 QUOTA SAMPLING

Quota sampling is the process of selecting a sample based on required, exact numbers, or quotas, of individuals or groups with varying characteristics. It is most often used in wide-scale survey research when listing all members of the population of interest is not possible. When quota sampling is involved, data gatherers are given exact characteristics and quotas of persons to be interviewed (e.g., 35 working women with children under the age of 16, 20 working women with no children under the age of 16). Obviously, when quota sampling is used, data are obtained from easily accessible individuals. Thus, people who are less accessible (e.g., more difficult to contact, more reluctant to participate) are underrepresented.



5.6.4 SNOWBALL SAMPLING

Chain goes until we get a amount of need

Selecting a few people who fit a researchers needs, then using those participants to identify additional participants, and so on, until the researcher has a sufficient number of participants. (Snowballing is most useful when it is difficult to find participants of the type needed.)

5.7 ADVANTAGES OF SAMPLING

1. Reduced Cost:

If the data are collected for the entire population, cost will be very high. It is economical of cost when the data are collected from a sample which is only a fraction of the population.

2. Greater Speed:

The use of sampling is economical of times also. Sampling is less time consuming than the census technique. Tabulation, analysis, etc. also take much less time in the case of a sample than in the case of a population.

3. Great Scope:

Complete enumeration of all units of the population are not only most often impracticable, but they require highly trained personnel and sophisticated equipment. Sample simplifies things and personnel with a little training can collect and handle the data. There is greater scope and flexibility of studies when a sample is used.

4. Greater Accuracy:

Sampling ensures completeness and a high degree of accuracy due to a limited area of operation. In dealing with a sample the volume of work is reduced, therefore careful execution of field work is possible. The processing of the data is also done more accurately, which in turn produces better results.

5. Organization of Convenience:

Sampling involves very few organizational problems. Due to small numbers, it does not require vast facilities. It is economical in respect of resources. The space and equipment required for this study are very small.

6. Intensive and Exhaustive Data:

As the number is limited, therefore it is possible to collect intensive and exhaustive data.

7. *Suitable in limited resources:*

In every society there are more problems and less resources, particularly when the people are poor and problems uncountable. This is the method which enables the researcher to work even with limited resources.

8. *Better Rapport:*

Usually it is difficult for the researcher to establish rapport with the respondents when they are large in number and many studies fail for want of good rapport between the researcher and the respondents. As the population of the study increases, the problem of rapport also increases. But in a manageable sample it is possible for the researcher to establish this meaningful rapport.

9. In a small sample, it becomes possible to scrutinize the data collected.

5.8 DISADVANTAGES OF SAMPLING

1. *Chances of Bias:*

The most common argument against the sampling method is the fact that it may involve biased selection and thereby lead us to draw erroneous conclusions. A bias is the sample may be caused either by faulty method of selection of individuals for the sample or the nature of the phenomena itself.

2. *Difficulties in selecting a truly representative sample:*

The results of a sample are accurate ad usable only when the sample is representative of the whole group.

Selection of a truly representative sample is very difficult particularly when the phenomena under study are of a complex nature. A number of reasons stand in the way of selecting good samples.

3. Need for specialized Knowledge:

The sampling method cannot be used by everybody. It requires a specialized knowledge in sampling technique, statistical analysis and calculation of probable error. In the absence of such knowledge, the researcher may commit serious mistakes.

4. Changeability of Units:

If the units of the population are not homogenous, the sampling technique will be unscientific. Although the number of cases is small, it is not always easy to stick to the selected cases especially in social sciences research. The cases of the sample may be widely dispersed. Some of them may refuse to cooperate with the researcher, and some others may be inaccessible. Because of these difficulties, all the cases may not be taken up. Sometimes the selected cases may have to be replaced by other cases. All this introduces a change in the stipulated subjects to be studied.

5. Impossibility of Sampling:

Sometimes the universe is too small, or too heterogeneous so that it is not possible to derive a representative sample. In such cases census study is the only alternative. In the studies, where a very high standard of accuracy is expected, the sampling method may be unsuitable. Even if the sample is drawn most carefully, there are always some chances of error.

Importance of Sampling

The sampling method was used in social sciences research as early as in 1754 by A. L. Bowley. Since then the method is increasingly being utilized. This technique has now made many a studies possible.

1. When the population is very large, it can be satisfactorily covered through sampling.
2. It saves a lot of time, energy and money.
3. Especially when the units of an area are homogeneous, sampling technique is really useful.
4. When the data are unlimited, the use of this method is really useful.
5. When cent percent accuracy is not required, the use of this technique becomes inevitable.
6. When the number of individuals to be studied is manageable, intensive study becomes possible.

CHARACTERISTICS OF A GOOD SAMPLE

1) A good sample is one which, within restrictions imposed by its size, will reproduce the characteristics of the population with the greatest possible accuracy.
2) It should be free from error due to bias or due to deliberate selection of the unit of the sample.

3) It should be free from random sampling error. It should not be selected by a procedure where there is a connection between the method of selection and the characteristics under consideration.

4) There should not be any substitution of originally selected unit by some other more convenient in any way.

5) It should not suffer from incomplete coverage of the units selected for study, i.e. it should not ignore the failures in the sample in responding to the study.

6) Relatively small sample properly selected may be much more reliable than large samples poorly selected. But at the same time it is very essential that the sample is adequate in size so that it can become really reliable.

7) In the samples only such units should be included, which as far as possible, are independent.

8) While constructing a sample, it is important that measurable or known probability sample techniques are used. This will substantially reduce the likely discrepancies.

CHAPTER – 6

TYPES OF INSTRUMENTS

There are various instruments available which yield a wide variety of data for a wide variety of purposes. Selection of an instrument for a particular research involves identification and selection of the most appropriate one from among alternatives. The instruments of education research are discussed as under:

6.1 TEST

A test is a means of measuring the knowledge, skill, feeling, intelligence, or aptitude of an individual or group.

Test objectivity means that an individual's score is not affected by the person scoring the test.

Validity is the most important quality of any test. Validity is concerned with what a test measures and for whom it is appropriate, reliability refers to the consistency with which a test measures whatever it measures.

Specification of conditions of administration, directions for scoring, and guidelines for interpretation are important characteristics of standardized tests.

TYPES OF TEST

A) Achievement Tests:

Achievement tests measure the current status of individuals with respect to proficiency in given areas of knowledge or skill.

Standardized achievement tests are available for individual curriculum areas such as reading and mathematics, and also in the form of comprehensive batteries that measure achievement in several different areas.

B) Aptitude Tests:

PreTest / Placement.

Aptitude tests are used to predict how well someone is likely to perform in a future situation.

Tests of general aptitude are variously referred to as scholastic aptitude tests, intelligence tests, and tests of general mental ability.

General Aptitude:

General aptitude tests typically ask the individual to perform a variety of verbal and nonverbal tasks that measure the individual's ability to apply knowledge and solve problems.

While general aptitude tests are intended to be measures of innate ability or potential, they appear actually to measure current ability.

Since they seem to do a reasonable job of predicting future academic success, and are in the sense measures of "potential" they are very useful to educators.

Specific Aptitude:

Specific aptitude tests attempt to predict the level of performance that can be expected of an individual following future instruction or training in a specific area or areas.

Aptitude tests are available for a number of specific areas such as algebra, music, and mechanical ability.

Multi-aptitude batteries, which measure aptitudes in several related areas, are available for the assessment of both academic and nonacademic aptitudes.

C) Personality Tests

Tests of personality are designed to measure characteristics of individuals along a number of dimensions and to assess feelings and attitudes toward self, others, and a variety of other activities, institutions, and situations. They may well be the most used tests in education research.

6.2 THE QUESTIONNAIRE

It is a data-gathering instrument through which respondent, answer questions or respond to statements in writing. It consist of series of questions or statements that are submitted to those from which information is desired.

TYPES OF QUESTIONNAIRE

Data in Numeric form.

1. Closed Form

I TOS
Rating scale. We easily Analyse

The questions that calls for short, check mark responses are known as the restricted or closed type. It is easy to fill out, takes less time and is fairly easy to tabulate and analyze. It is more acceptable and convenient for the respondents.

2. Open Form:

It calls for a free response in the respondents own words. The respondent frames and supplies his own response. It provides for greater depth of response. It is difficult to interpret, tabulate and summarize.

ADVANTAGES:

1. Economical:

It is economical way of collecting informations of significance to researcher. It is economical in term of time, effort and cost.

2. High Coverage:

It is the best tool for scattered subjects. It permits a nation wide as well as international coverage. It makes possible contact with those who could not otherwise be reached.

3. Group administration:

It allows group administration and covers a large group at the same time.

4. Ease:

It is easy to plan, construct and administer.

5. Dependability:

It is regarded as dependable when used to obtain statement of facts.

6. Personal informations:

Personal informations can be obtained when respondents are allowed to omit name and signature or if assured that their replies will be kept confidential.

7. Responses at leisure time:

It permits the respondents to answer at leisure time. It does not place pressure for immediate response.

8. Validity of responses:

The written responses in the language of respondents increases the validity of responses. In interview or observation, recording by the researcher decreases the validity of responses.

9. Great Potentiality:

The questionnaire has great potentialities when it is used properly. If it is eliminated, the progress in many areas of education would be greatly handicapped. It is easy to tabulate and analyze.

LIMITATIONS

1. Low Responses:

The mailed questionnaires have low response rates.

2. Limited validity of data:

If the response rate of mailed questionnaires is less than 40%, the data obtained are of limited validity.

3. No clarification:

In case of interview, clarification can be made of ambiguous questions but in questionnaire there is no provision of clarification.

4. Emotional topics:

It is not helpful in finding informations about complex emotional subjects. People do not like to put their personal feelings in black and white. The delicate and controversial issues can be discussed only in interview.

5. Body language remains unnoticed:

The behaviours, gestures, emotions and reactions remain unnoticed.

6. Lack of Motivation:

The questionnaire has no potential to motivate the subjects to respond on a specific issue whereas in interview, interviewer can motivate the subjects and can get responses on a specific issue.

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7. Lacks representative sample:

The respondents who return the questionnaires may not constitute a representative sample. Only more responsible, research minded or those in favour of issue may choose to respond.

8. It is impossible to put down certain delicate issues in writing

9. The questionnaire can not be used will illiterate subjects and children.

6.3 INTERVIEW:

Interview is a technique of collecting information orally form others in a face to face situation. It is a two way method which permits an exchange of ideas and information. The interviewer asks a series of open ended probing questions. People are more willing to talk than write. It can produce in depth data which is not possible with a questionnaire, although it is expensive and time consuming.

TYPES OF INTERVIEW:

There are may types of interviews depending upon purpose, nature and scope. These types are as under:

1. Individual Interview:

It is an interview conducted with one person at a time. The subject feel free to express himself fully and truth fully.

2. Group Interview:

It is an interview conducted with many persons at a time. They can help one another recall, verify or rectify informations. They can present wide range of informations and varied view points.

3. Single Interview:

When interview is held by one interviewer it is single interview.

4. Panel Interview:

When interview is conducted by many interviewers, it is panel interview. This panel may consist of various experts related to different fields e.g; subject specialists, administrator, psychologist, social worker etc.

5. Structured Interview:

Eg Unchangle.
Questions are planned

In structured interview preplanned questions are asked in a specific sequence. The interviewer follows the rigid order. There is no flexibility of asking new or situation oriented questions. different persons have different Answer.

6. Unstructured Interview:

The Unstructured interview are informal. The questions being asked are not planned. The interviewer asks questions according to the situation. Unstructured interviews are flexible.

7. Non-directive Interviews:

It is indepth interview in which interviewer permits subject to talk free to take insight into nature of attitudes, motives, feelings and beliefs. It is psychoanalytical in nature. It is used to take insight into hidden motives.

8. *Focused Interview:*

In focused interview, the respondent is allowed to express himself completely but the interviewer directs the line of thought. It focuses attention on specific experiences. It confines the conversation to the relevant issues.

ADVANTAGES OF INTERVIEW

1. *High Responses:*

The percentage of response is likely to be much higher than in case of questionnaire.

2. *Greater depth:*

It permits greater depth and provides a true picture of opinions and feelings.

3. *Flexibility:*

The interview is more flexible than questionnaire. The interviewer can elaborate and clarify the questions when the respondent is unable to understood.

4. *Establishment of Rapport:*

The interview permits establishment of greater rapport and stimulates respondent to give complete and valid answer. By establishing rapport, interviewer can get delicate and confidential informations.

5. *Assurance of Confidentiality:*

The interviewer can give assurance that the facts will be kept confidential and will be used properly.

6. Two-way Communication:

It permits exchange of ideas and thoughts. It gives opportunity for give and take.

7. No doubt or misunderstanding:

The interviewer is personally present to remove the doubt or misunderstanding.

8. Friendly Atmosphere:

The interviewer can create a friendly atmosphere which is necessary for obtaining desired data.

9. High Reliability:

The data collected through interviews has been found to be more reliable.

10. Only method of specific categories:

It is the only method used for categories like young children, illiterates, abnormal children and mentally retarded children.

LIMITATIONS OF INTERVIEW

1. Expensive:

Interview is comparatively a costly affair. The cost per interview is higher than any technique used to collect data.

2. Bias:

Interviewer bias is the major weakness of the interview. Interviewers tend to get data that aggress with their personal conviction.

3. *Time Consuming:*

A busy person may prefer to fill out a questionnaire at convenient time rather than submit to a long interview.

4. *Large no of Workers needed:*

A large no. of trained field workers are required to take interview which entails a lot of expenditures.

5. *Recording of data is difficult:*

The data obtained from interview is difficult to record. Writing during interview affect rapport and recording affect conduct of interview. Writing after interview leads toward unconscious selection of the material.

6. *Does not suit specific categories:*

Interview does not suit infants, sky people, and deaf people.

7. *Needs Expertise:*

As the objectivity, sensitivity, and insight of the interviewer is crucial, high level of expertise is required to conduct an interview.

8. *Tape recording makes the interviewee cautious:*

Cautions take him away from giving any secrets.

6.4 OBSERVATION

Observation is more natural way of assessing children. Observation seeks to ascertain what children think and do by watching them in action as they express themselves in various situations and activities. Direct observation has come to be looked upon as a scientific procedure. It is the first procedure of science. All scientific data must originate in some experience or perception.

CHARACTERISTICS OF OBSERVATION

1) Scientific observation is systematic.
2) Observation is specific.
3) Scientific observation is objective.
4) Scientific observation is quantitative.
5) Observation is recorded immediately.
6) Observation is expert.
7) Observation is verifiable.

TYPES OF OBSERVATION

1. *Non-Participant Observation*

In non-participant observation, the researcher is not directly involved in the situation to be observed.

2. *Participant Observation*

In participant observation, the researcher is directly involved in a situation to be observed.

3. *Naturalistic Observation*

In naturalistic observation, the researcher does not control or manipulate anything. He works hard at not affecting the observed situation.

4. Simulation Observation

In simulation observation, the observer creates the situation to be observed. In it observer observes behaviour that occurs infrequently in natural setting e.g. role playing.

5. Structured Observation

In structured observation the observer observes what is already planned to observe. It is a formal observation in which observer observes restricted aspects of the situation.

6. Unstructured Observation

It is non-formal observation which is not preplanned. It allows maximum flexibility in observation to get a true picture of the phenomenon as a whole.

MERITS

1. Reliable and Objective

Observation is more reliable and objective as actual behaviour of the child is recorded.

2. Natural Setting

It is the study of an individual in a natural setting so it is more useful than the restricted study in test situation.

3. Useful for Young and Shy Children

This method is very useful for young and shy children. It is easier to observe the young children rather than taking information by interview.

4. No Need of Equipment or Tool

This method does not require special equipment or tools. It can be used by all teachers with a little training or experience.

5. Useful for Individuals as well as Groups

The method of observation is adaptable both to individuals and groups.

6. Immediate Detection of Problems

The problems can be immediately detected and corrective actions can be taken immediately.

LIMITATIONS

1. Subjectivity

There is a great chance for personal prejudices and bias of the observer.

2. Less accuracy

As observation is recorded after some time of occurrence of events so records written may not be 100% accurate.

3. Interference

Observation is self interfering. The act of observing produces distortion of the phenomenon.

4. Objective Recording

There is danger of concentrating on the aspects of limited significance simply because they can be recorded objectively and accurately.

5. Lack of Competency

Lack of competency of observer may hamper validity and reliability of observation.

6. Unnatural Situation

The people being observed become conscious and begin to behave in an unnatural manner. The situation does not remain real and natural.

7. Expensive

It is a costly affair. It involves expenses on traveling, staying and purchasing of sophisticated equipment.

8. Slow and Laborious

It is slow and laborious process. A particular situation desired to be observed might not occur for long time.

6.5 RATING SCALE:

Mostly used in Observation & Questionnaire.

The rating scale involves qualitative description of a limited number of aspects of a thing or of traits of a person. The classifications may be set up in five to seven categories in such terms as: 5 to 7.

1	superior	above average	Average	fair	inferior
2	excellent	good	Average	below average	Poor
3	always	frequently	Occasionally	rarely	Never

6.6 ATTITUDE SCALES

Attitude scales attempt to determine what an individual believes, perceives, or feels. Attitudes can be measured toward self, others, and a variety of other activities, institutions, and situations. There are four basic types of scales used to measure attitudes: Likert scales, semantic differential scales, Thurston scales, and Guttman scales. The first two are used more often.

Most used scale.

i. A Likert Scale:

A Likert scale asks an individual to respond to a series of statements by indicating whether she or he strongly agrees (SA), agrees (A), is undecided (U), disagrees (D), or strongly disagrees (SD) with each statement. Each response is associated with a point value, and an individual's score is determined by summing the point values for each statement. For example, the following point values might be assigned to responses to positive statements. SA = 5, A = 4, U = 3, D = 2, SD = 1.

For negative statements, the point values would be reversed, that is SA = 1, A = 2 and so on. An example of a positive statement might be "Short people are entitled to the same job opportunities as tall people." A high point value on a positively stated item would indicate a positive attitude and a high total score on the test would be indicative of a positive attitude

ii. A semantic differential scale:

A semantic differential scale asks an individual to give a quantitative rating to the subject of the attitude scale on a number of bipolar adjectives such as good-bad, friendly unfriendly, positive-negative. The respondent indicates the point on the continuum between the extremes that represents her or his attitude.

Most adjective pairs represent one of three dimensions which they labeled evaluation (e.g., good-bad) potency (e.g., strong-weak), and activity (e.g., active-passive). For example on a scale concerning attitudes toward property taxes, the following items might be included.

Necessary ---------------------------------- Unnecessary

Fair -------------------------------- Unfair

Each position on a continuum has an associated score value; by totaling score values for all items, it can be determined whether the respondent's attitude is positive or negative. Semantic differential scales usually have 5 to 7 intervals with a neutral attitude being assigned a score value of 0. For the above items, the score values would be a follows.

Necessary ---------------------------------- Unnecessary

3 2 1 0 -1 -2 -3

Fair -------------------------------- Unfair

3 2 1 0 -1 -2 -3

A person who checked the first interval (i.e., 3) on both of these items would be indicating a very positive attitude toward property taxes.

iii. A Thurstone Scale:

A Thurstone scale asks an individual to select from a list of statements that represent different points of view from those with which he or she is in agreement. Each item has an associated point value between 1 and 11; point value for each item are determined by averaging the values of the items assigned by a number of "judges". An individual's attitude score is the average point value of all the statements checked by that individual.

iv. A Guttman scale

A Guttman scale also asks respondents to agree or disagree with a number of statements. A Gutman scale, however, attempts to determine whether an attitude is unidimensional if it produces a cumulative scale. In a cumulative scale, an individual who agrees with a given statement also agrees with all related preceding statements; for example, if you agree with statement 3, you also agree with statement 2 and 1.

6.7 PROJECTIVE DEVICES

A projective instrument enables subjects to project their internal feelings, attitudes, needs, values, or wishes to an external object. Thus the subjects may unconsciously reveal themselves as they react to the external object. The use of projective devices is particularly helpful in counteracting the tendency of subjects to try to appear in their best light, to respond as they believe they should.

Projection may be accomplished through a number of techniques:

i. Association:

The respondent is asked to indicate what he or she sees, feels, or thinks when presented with a picture, cartoon, ink blot, word, or phrase. The Thematic Apperception Tests, the Rorschach Ink Blot Test, and various word-association tests are familiar examples.

ii. Completion:

The respondent is asked to complete an incomplete sentence or task. A sentence-completion instrument may include such items as:

My greatest ambition is To became a successful woman.

My greatest fear is ------------------------

I most enjoy with family & friends.

I dream a great deal about ----------------

I get very angry when --------------------

iii. Role-playing:

Subjects are asked to improvise or act out a situation in which they have been assigned various roles. The researcher many observe such traits as hostility, frustration, dominance, sympathy, insecurity, prejudice – or the absence of such traits.

iv. Creative or constructive:

Permitting subjects to model clay, finger paint, play with dolls, play with toys, or draw or write imaginative stories

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The respondent is asked to indicate what he or she sees, feels, or thinks when presented with a picture, cartoon, ink blot, word, or phrase. The Thematic Apperception Tests, the Rorschach Ink Blot Test, and various word-association tests are familiar examples.

ii. Completion:

The respondent is asked to complete an incomplete sentence or task. A sentence-completion instrument may include such items as:

My greatest ambition is to became a succesful woman.

My greatest fear is ------------------------

I most enjoy with family & friends.

I dream a great deal about ----------------

I get very angry when ---------------------

iii. Role-playing:

Subjects are asked to improvise or act out a situation in which they have been assigned various roles. The researcher many observe such traits as hostility, frustration, dominance, sympathy, insecurity, prejudice – or the absence of such traits.

iv. Creative or constructive:

Permitting subjects to model clay, finger paint, play with dolls, play with toys, or draw or write imaginative stories about assigned situations may be revealing. The choice of color, form, words, the sense of orderliness, evidence of tensions, and other reactions may provide opportunities to infer deep-seated feelings.

6.8 CHECKLIST:

The checklist, the simplest of the devices, is a prepared list of behaviour or items. The presence or absence of the behaviour may be indicated by checking yes or no, or the type or number of items may be indicated by inserting appropriate word or number.

Trait	Absent / Present
Regularity	______________
Cleanliness	______________
Consistency	______________
Classroom involvement	______________

6.9 MEASUREMENT SCALES

Measurement scales are of following types:

6.9.1 NOMINAL SCALES

A nominal scales represents the lowest level of measurement. **Such a scale classifies persons or objects into two or more categories.** Whatever the basis for classification, a person can only be in one category, and members of a given category have a common set of characteristics. Categories may be referred to as true or artificial. True categories are categories into which persons or objects naturally fall, independently of the research study.

True Categories Having some Naturally Based Traits

Variables which involve true categories include:

1) Sex (female versus male)
2) Number of siblings (0 versus 1 verses 2 verses 3, etc.)
3) Employment status of mother (full-time verses part-time verses unemployed)
4) Preschool attendance (attended preschool verses did not attend preschool)
5) Type of school (e.g., public verses private)

Artificial categories

Artificial categories are categories which are operationally defined by the researcher. Variables which require artificial categories include:

1) Height (e.g., tall verses short)
2) Trait anxiety (e.g., high anxious verses low anxious)
3) IQ (e.g., high IQ verses average IQ verses low IQ)
4) Family income (e.g., above average verses below average)
5) Learning style (e.g., field independent verses field dependent)

6.9.2 ORDINAL SCALES

Ranking

An ordinal scale not only classifies subjects but also ranks them in terms of the degree to which they possess a characteristic of interest. In other words, an ordinal scale puts the subjects in order from highest to lowest, from most to least. With respect to height, for example, 50 subjects might be ranked from 1 to 50; the subject with rank 1 would be the tallest and the subject with rank 50 would be the shortest. It would be possible to say that one subject was taller or shorter than another subject. Indicating one's rank in a graduating class, or expressing one's score as a percentile rank, are measures of relative standing that represent ordinal scales.

Although ordinal scales to indicate that some subjects are higher, or better, than other, they do not indicate how much higher or how much better. In other words, intervals between ranks are not equal; the difference between rank 1 and rank 2 is not necessarily the same as the difference between rank 2 and rank 3, as the example below illustrates:

Rank	Height
1	6'2"
2	6'1"
3	5'11"
4	5'7"
5	5'6"
6	5'5"
7	5'4"
8	5'3"
9	5'2"
10	5'0"

The difference in height between the subject with rank 1 and the subject with rank 2 is 1 inch; the difference between rank 2 and rank 3 is 2 inches. In the example given, differences in height represented by differences in rank range from 1 inch to 4 inches. Thus, while an ordinal scale results in more precise measurement than a nominal scale, it still does not allow the level of precision usually desired in a research study.

6.9.3 INTERVAL SCALES

An interval scale has all the characteristics of a nominal scale and a ordinal scale, but in addition it is based upon predetermined equal intervals. Most of the tests used in educational research, such as achievement tests, aptitude tests, and intelligence tests, represent interval scales. Therefore, you will most often be working with statistics appropriate for interval data. When we talk about "scores", we are usually referring to interval data. When scores have equal intervals it is assumed, for example, **that the difference between a score of 30 and a score of 40 is essentially the same as the difference between a score of 50 and score of 60, and the difference between 81 and 82 is approximately the same as the difference between 82 and 83.** Interval scales, however, do not have a true zero point. Such scales typically have an arbitrary maximum score and an arbitrary minimum score, or zero point. If and IQ test produces scores ranging from 0 to 200, a score of 0 does not indicate the absence of intelligence, nor does a score of 200 indicate possession of the ultimate intelligence. A score of 0 only indicates the lowest level of performance possible on that particular test and a score of 200 represents the highest level. Thus, scores resulting from administration of an interval scale can be added and subtracted, but not multiplied or divided. We can say that an achievement test score of 90 is 45 points higher than a score of 45, but we cannot say that a person scoring 90 knows twice as much as a person scoring 45. Similarly, a person with a measured IQ of 140 is not necessarily twice as smart, or twice as intelligent, as a person with a measured IQ of 70. For most educational measurement, however, such generalizations are not needed.

RATIO SCALES

A ratio scale represents the highest, most precise, level of measurement. A ratio scale has all the advantages of the other types of scales and a addition it has a meaningful, true zero point. Height, weight, time, distance, and speed are examples of ratio scales. The concept of "no time", for example, is a meaningful one. Because of the zero point, not only can we say that the difference between a height of 3'2" and a height of 4'2" is the same as the difference between 5'4" and 6'4", but also that a man 6'4" is twice as tall as a child 3'2". Similarly, 60 minutes is 3 times as long as 20 minutes, and 40 pounds is 4 times as heavy as 10 pounds. Thus, with a ratio scale we can say that Frankenstein is tall and Igor is short (nominal scale), Frankenstein is taller than Igor (ordinal scale), Frankenstein is 7 feet tall and Igor is 5 feet tall (interval scale), and Frankenstein is seven-fifths as tall as Igor. Since most physical measures represent ratio scales, but not psychological measures, ratio scales are not used very often in educational research.

6.10 CRITERION FOR THE SELECTION OF INSTRUMENTS

Validity of Measuring Instruments

1) Validity is the degree to which a test measures what it is supposed to measure, thus permitting appropriate interpretations of test scores.

2) A test is not valid per se; it is valid for a particular interpretation and for a particular group. Each intended test use requires its own validation. Additionally, validity is measured on a continuum—tests are highly valid, moderately valid, or generally invalid.

3) Content validity assesses the degree to which a test measures an intended content area. Content validity is of prime importance for achievement tests.

4) Content validity is determined by expert judgment of item and sample validity, not by statistical means.

5) Criterion-related validity is determined by relating performance on a test to performance on a second test or other measure.

6) Criterion validity has two forms, concurrent and predictive. Concurrent validity is the degree to which the scores on a test are related to scores on another test administered at the same time or to another measure available at the same time.

Predictive validity is the degree to which scores on a test are related to scores on another test administered in the future. In both cases, a single group must take both tests.

7) Construct validity is a measure of whether the construct underlying a variable is actually being measured.

8) Construct validity is determined by a series of validation studies that can include content and criterion-related approaches. Both confirmatory and disconfirmatory evidence are used to determine construct validity.

9) Consequential validity is concerned with the potential of tests to create harmful effects for test takers. This is a new but important form of validity.

10) The validity of any test or measure can be diminished by such factors as unclear test directions, ambiguous or difficult test items, subjective scoring, and nonstandardized administration procedures.

Reliability of Measuring Instruments

1) Reliability is the degree to which a test consistently measures whatever it measures. Reliability is expressed numerically, usually as a coefficient ranging from 0.0 to 1.0; a high coefficient indicates high reliability.

2) Measurement error refers to the inevitable fluctuations in scores due to person and test factors. No test is perfectly reliable, but the smaller the measurement error, the more reliable the test.

3) The five general types of reliability are stability, equivalence, equivalence and stability, internal consistency, and scorer/rater.

4) Stability, also called test-retest reliability, is the degree to which test scores are consistent over time. It is determined by correlating scores from the same test, administered more than once.

5) Equivalence, also called equivalent-forms reliability, is the degree to which two similar forms of a test produce similar scores from a single group of test takers.

6) Equivalence and stability reliability is the degree to which two forms of a test given at two different times produce similar scores as measured by correlations.

7) Internal consistency deals with the reliability of a single test taken at one time. It measures the extent to which the items in the test are consistent among themselves and with the test as a whole. Split-half, Kuder-Richardson 20 and 21, and Cronbach's alpha are the main approaches to obtaining internal consistency.

8) Split-half reliability is determined by dividing a test into two equivalent halves (e.g., odd items vs. even items), correlating the two halves, and using the Spearman-Brown formula to determine the reliability of the whole test.

9) Kuder-Richardson reliability deals with the internal consistency of tests that are scored dichotomously (i.e., right, wrong), whereas Cronbach's alpha deals with the internal consistency of tests that are scored with more than two choices (agree, neutral, disagree or 0, 1, 2, 3).

10) Scorer/rater reliability is important when scoring tests that are potentially subjective. Interjudge reliability refers to the reliability of two or more independent scorers, whereas intrajudge reliability refers to the reliability of a single individual's ratings over time.

Reliability Coefficients

1) The acceptable level of reliability differs among test types, with standardized achievement tests having very high reliabilities and projective tests having considerably lower reliabilities.

2) If a test is composed of several subtests that will be used individually in a study, the reliability of each subtest should be determined and reported.

Standard Error of Measurement

1) The standard error of measurement is an estimate of how often one can expect test score errors of a given size. A small standard error of measurement indicates high reliability; a large standard error of measurement indicates low reliability.

2) The standard error of measurement is used to estimate the difference between a person's obtained and true scores. Big differences indicate low reliability.